

WASTEWISE SCAN · REQUIREMENTS

WasteWise Scan requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WSC-REQ-001	Version	0.2
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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/wastewise-scan		
Source	docs/03-requirements.md @ d904c54		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise Scan requirements

These are first-pass requirements for the concept. Targets are proposals for review, to be checked by calculation at TRL 3 and revised after co-design with users. Status is the concept estimate from WSC-PRC-001; **R1 and R12 are not met or not shown to be met**, and R3 and R10 are unverified.

Table 1. Requirements.

ID	Requirement	Target	Concept status	Verification (TRL 3 or later)
R1	Identify common resins	PET, HDPE, PP, PS and PVC, plus "other" and "unknown"; 95 % or more correct on clean, dry, non-black items of 1 mm wall or more	Not shown. Plausible from prior work with discrete NIR bands; PE versus PP near the 1,700 nm band edge is the main risk	Band selection study on published reference spectra; later a labeled item set of 200 or more
R2	Fast result	1.5 s or less from button press to result	Met by estimate (about 0.6 s)	Timing budget
R3	Fail safe on unreadable items	Black, dark, wet or very thin items read "unknown"; wrong resin shown for 2 % or fewer of all scans	Unverified; black items read unknown by design	Confidence threshold study; later item test
R4	Work in sunlight	Same result indoors and in direct sun up to 100 klx when the shroud is pressed to the item	Met by design (shroud plus dark-frame subtraction), unverified for curved items	Stray light estimate
R5	Last a shift	8 h or more at 4 scans per minute on one charge; USB-C charging	Met by estimate (about 17 h)	Power budget
R6	One-handed and light	250 g or less; body no larger than 170 x 65 x 40 mm (6.7 x 2.6 x 1.6 in) plus shroud	Met by estimate (about 170 g; 160 x 62 x 34 mm)	Massing model, then weighing
R7	Readable by anyone	Result as resin code, name, color and icon, legible at arm's length in direct sun; local language; no reading needed to use	Met by design with a 1.9 in IPS display; sunlight legibility unverified	Design review with users
R8	Local price grade	Shows a price grade from a table the user or cooperative edits on the device; works offline	Met by design	Design review
R9	Log scans	10,000 or more scans stored on the device with time, result, confidence and spectrum; export as CSV over USB; no cloud needed	Met by estimate (about 64 B per scan, under 1 MB)	Storage calculation
R10	Rugged	Survives 1.2 m (4 ft) drop onto concrete; IP54 target; 0 to 45 °C	Unverified	Design review, later drop and spray tests
R11	Stay calibrated	White reference check in 10 s or less using the calibration cap; warns when a check is overdue or drift exceeds 5 %	Met by design	Design review
R12	Low cost	Parts cost \$150 or less per unit; no custom PCB for the first build	Not met (about \$163, 9 % over)	Priced BOM
R13	Work with WasteWise-ml	Low-confidence scans hand off to WasteWise-ml; scan records use a shared, documented format	Met by design, format not yet defined	Interface note with WasteWise-ml
R14				

ID	Requirement	Target	Concept status	Verification (TRL 3 or later)
	Eye-safe illumination	LED emission in the IEC 62471 exempt group; LEDs pulse only during a scan	Met by estimate (a few mW per LED, under 1 s)	Emission estimate from datasheets

■ Assumptions

- "Clean and dry" means rinsed or wiped items without labels over the scan spot. Real waste is dirtier; the classifier must reject rather than guess.
- About 10 to 15 % of items in a mixed post-consumer stream are black or very dark (estimate; varies strongly by region and should be measured with users).
- Four scans per minute is a busy spot-check pace, not every item: most items are sorted by eye and the scanner settles doubtful ones.
- Resin prices vary by city and season; the device stores grades (for example A, B, C, reject) and a local price list, not global prices.